

```
# Removes the existing version of the transformers library.
```

```
!pip uninstall -y transformers
# Installs a specific version (4.46.3) along with its required libraries.
!pip install -U transformers==4.46.3 \
    accelerate \
    sentencepiece \
    huggingface-hub==0.26.2
```

```
Found existing installation: transformers 4.46.3
Uninstalling transformers-4.46.3:
  Successfully uninstalled transformers-4.46.3
Collecting transformers==4.46.3
  Using cached transformers-4.46.3-py3-none-any.whl.metadata (44 kB)
Requirement already satisfied: accelerate in /usr/local/lib/python3.12/dist-packages (1.14.0)
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dist-packages (0.2.2)
Requirement already satisfied: huggingface-hub==0.26.2 in /usr/local/lib/python3.12/dist-packages (0.26.2)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (3.29.4)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (26.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (2025.11)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (2.32.4)
Requirement already satisfied: tokenizers<0.21,>=0.20 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (0.15.1)
Requirement already satisfied: safetensors>=0.4.1 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (0.8.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (4.67.3)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub==0.26.2) (2025.10.2)
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub==0.26.2) (4.12.2)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate) (5.9.5)
Requirement already satisfied: torch>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from accelerate) (2.11.0+cu128)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.1.6)
Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,
Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelera
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Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cubl
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit
Requirement already satisfied: nvidia-cuffrt-cu12==11.3.3.83.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cubl
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cubl
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Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[c
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Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->transformers=
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.46.3) (3.
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.46.
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.46.
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings<13,>=12.9.4->
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=2.0.0->
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=2.0.0->accelerate)
Using cached transformers-4.46.3-py3-none-any.whl (10.0 MB)
Installing collected packages: transformers
Successfully installed transformers-4.46.3
```

```
# Import Model and Tokenizer from transformer Library
from transformers import AutoModelForCausalLM, AutoTokenizer
# Load model(Phi-3) and tokenizer
model = AutoModelForCausalLM.from_pretrained(
    "microsoft/Phi-3-mini-4k-instruct",
    # Use the GPU
    device_map="cuda",
    # Choose the best data type automatically
    torch_dtype="auto",
    # Allow the model to run its own code
    trust_remote_code=
    True
    ,
)
# Load the tokenizer for the model
tokenizer = AutoTokenizer.from_pretrained("microsoft/Phi-3-mini-4k-instruct")
```

```

Downloading shards: 100%                2/2 [00:06<00:00, 3.26s/it]
model-00002-of-00002.safetensors: 100% 2.67G/2.67G [00:06<00:00, 228MB/s]
Loading checkpoint shards: 100%          2/2 [00:33<00:00, 15.77s/it]
generation_config.json: 100%             181/181 [00:00<00:00, 11.3kB/s]
tokenizer_config.json: 3.44k/? [00:00<00:00, 221kB/s]
tokenizer.json: 1.94M/? [00:00<00:00, 56.7MB/s]
added_tokens.json: 100%                  306/306 [00:00<00:00, 35.3kB/s]
special_tokens_map.json: 100%            599/599 [00:00<00:00, 60.0kB/s]

```

```

# Import pipeline , which Combines the model and tokenizer to generate the output
from transformers import pipeline
# Create a pipeline
generator = pipeline(
# Tells the pipeline to perform text generation task
"text-generation",
# Uses model that we are imported (Phi-3)
model=model,
# Uses the tokenizer that convert text into tokens and tokens to text =
tokenizer=tokenizer,
# return only newly generated text (Avoid the original prompt )
return_full_text=
False
,
# Allow the model to generate only 500 new tokens(words)
max_new_tokens=500,
# Tells the model to generate the specific(next) text (No randomness)
do_sample=
False
)

```

```

# The prompt (user input / query)
messages = [{"role": "user", "content": "Create a funny joke about chickens.\n"}
]
# Generate output
output = generator(messages)
print(output[0]["generated_text"])

```

Why did the chicken join the band? Because it had the drumsticks!